VACUTECH · PHASE 2 · BUSINESS INTELLIGENCE & DATA MINING

From RMAs to Recommendations

Three CRM questions, three transparent data-mining models

Course module: DLMDSEBA02 • Composite presentation

W1 · Apriori

306 rules from 7460 PM cases

Pareto: 26 SKUs cover ≈80%

W2 · Decision Tree

R^2 = 0.93 on 2000 test cases

MAE ≈ 1.07 days at intake

W3 · DT classifier

Threshold = 0.3, recall 49%

Accuracy 85% on 2000 test cases

All KPIs are read live from exports/metrics_summary.json

CRM tasks and the data-mining toolbox

Customer Relationship Management aims to identify, attract, retain and develop customers. Data mining supports each of these CRM tasks with a different family of algorithms[1)][14]. VacuTech's three W questions sit on the retention and development side: stock the right parts, make credible promises, and prevent rework before customers feel it.

Customer Identification

Segmentation, target selection

DM family: Clustering, classification

Customer Attraction

Direct marketing, lead scoring

DM family: Classification, association rules

Customer Retention

Service quality, churn, complaint triage

DM family: Classification (DT), regression

Customer Development

Cross-/up-sell, lifetime value, bundling

DM family: Association rules, regression

VacuTech mapping ● W1 → Customer Development (parts bundling) ● W2 → Customer Retention (credible ETAs) ● W3 → Customer Retention (QA-failure prevention)

[14] Ngai et al. 2009 [1] Shearer 2000

Which parts frequently co-occur and should be bundled together?

Business question

Technicians often consume additional parts that break together. We want to find which parts frequently co-occur across all repairs completely independently of PM, so planners can pre-stock or bundle them and replace them in every repair where one is flagged.

Data used

- 10,000 repair cases from repairs.csv (independent of PM flag)
- Add-on lines from parts_used.csv (kit_part_flag = 0)
- Pivoted into a boolean basket matrix: 10,000 cases x 36 add-on parts
- No scaling or imputation required — Apriori runs directly on baskets

Apriori association rules + Pareto

- Apriori (mlxtend) discovers frequent co-occurring add-on parts
- Rules ranked by lift, retained when lift exceeds independence
- Pareto chart on part counts complements rules with a SKU shortlist

Fixed hyperparameters

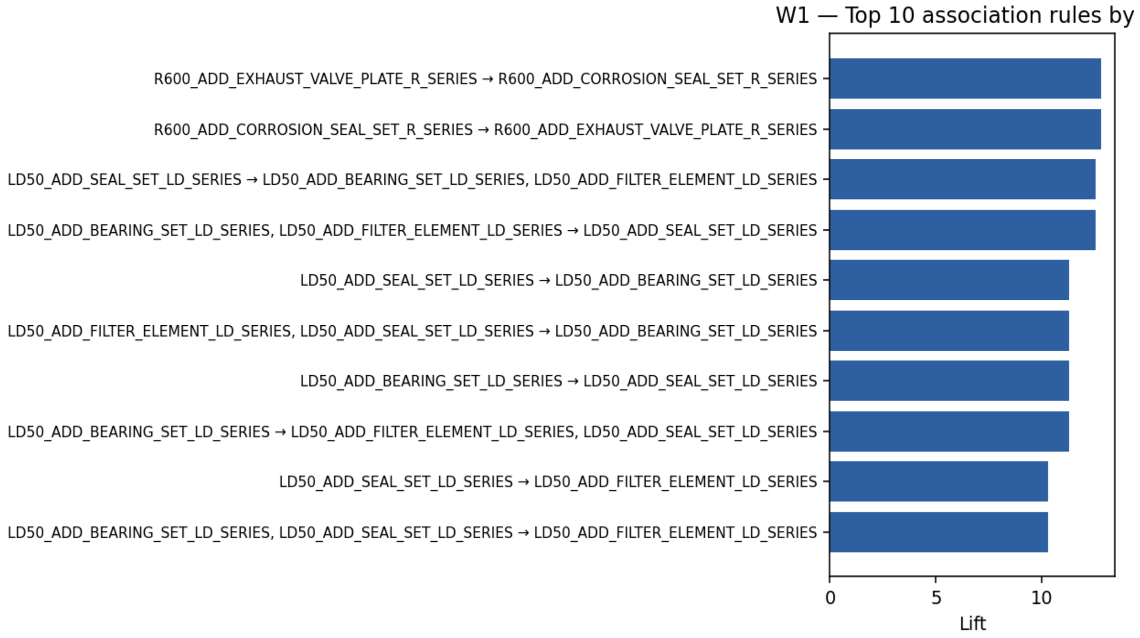
- min_support = 0.05 (≈ 25-50 baskets at this dataset size)
- min_lift = 1.2 (meaningfully stronger than chance)
- Outputs: w1_rules.csv, w1_metrics.json, two PNG charts

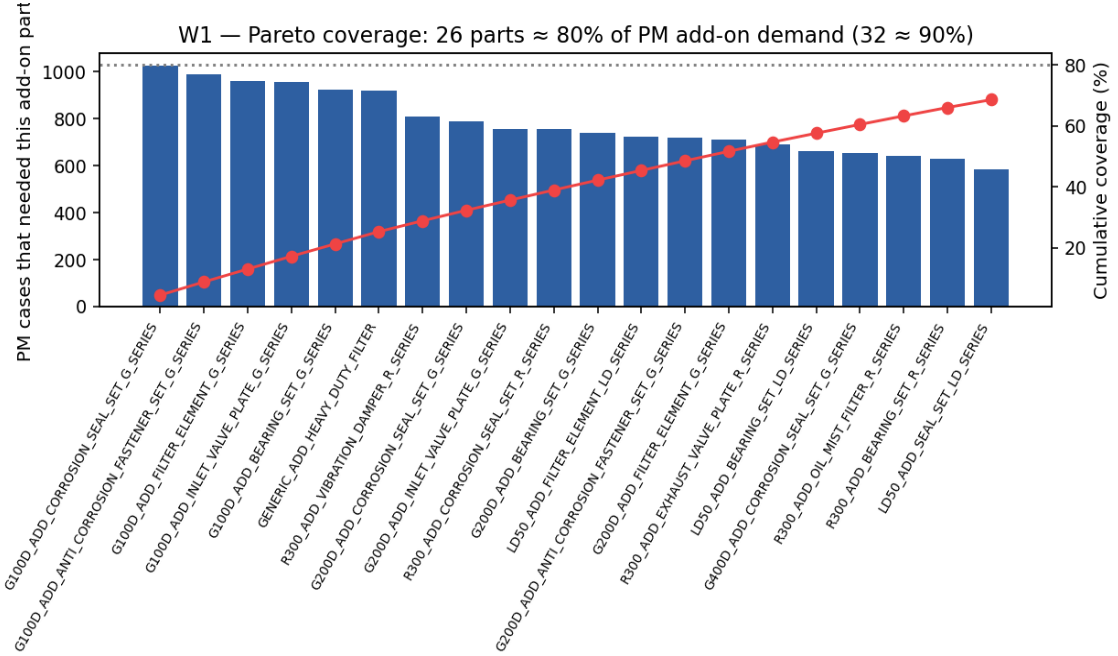
Frameworks

mlxtend.apriori · mlxtend.association_rules

[3] Agrawal & Srikant 1994 [4] Brin et al. 1997

Predictive stocking shortlist + direct bundling rules





Bundle Rules Found	Rule Strength (Lift)	Bundling Confidence	80% / 90% SKU Headcount
306	12.8x	84%	26 / 32 SKUs

Management action

- Stock the top 20 high-demand add-on parts at every regional warehouse to satisfy 80% of extra demand.
- Bundle complementary wear-and-tear parts (e.g. O-rings + Seals) directly into standard PM kits.
- Verify first-visit completion rates before vs. after roll-out to measure customer satisfaction gains.

[6] Wang et al. 2018 [7] Didriksen et al. 2026

At intake, how long will this repair take?

Business question

The service planner must commit to a reliable repair completion date. The prediction model must use only safe, intake-time features (no QA or completion fields) and stay fully explainable to a front-desk advisor.

Data used

- Target: Actual Repair Duration (in Days)
- Predictors: Pump age, technician experience, estimated parts cost
- Job Info: Specific pump model, failure category, complexity level, region
- Preparation: Clean incomplete records + format categorical groups

Decision Tree Regressor (Whitebox AI)

- Fully transparent method — logic splits readable on a whiteboard
- Decision paths trace exactly how job drivers add up to the ETA
- Evaluated on a completely separate, held-out test dataset

Fixed hyperparameters

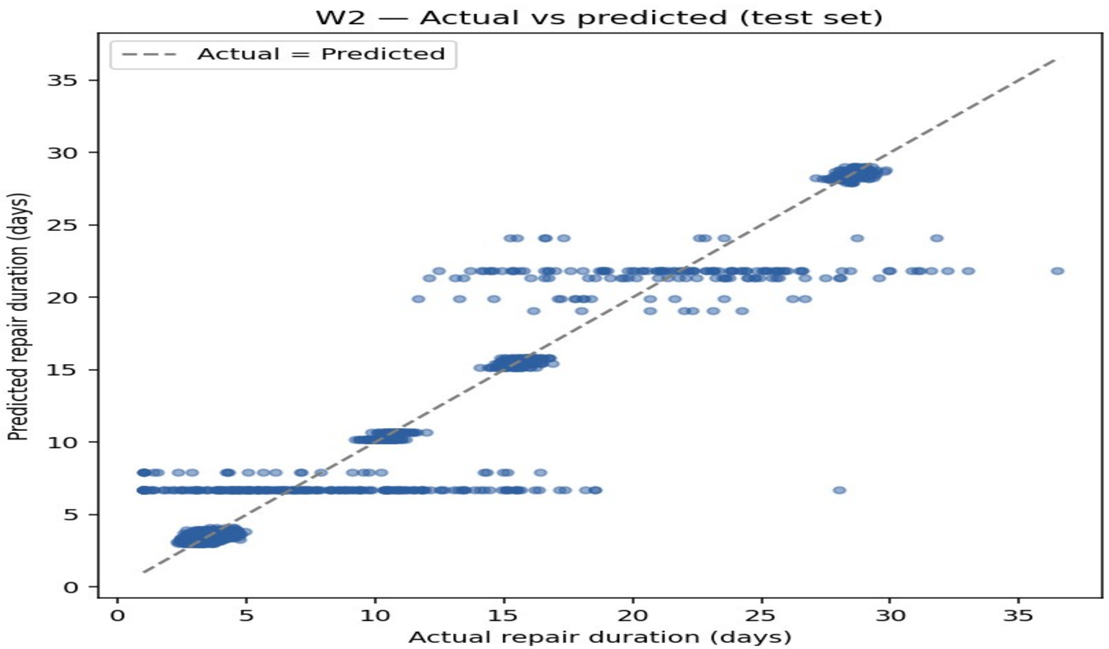
- max_depth = 6 (captures age x complexity interactions without over-fitting)
- min_samples_leaf = 20 (≥ 20 historic cases per decision path)
- random_state = 7 (fully repeatable and auditable models)

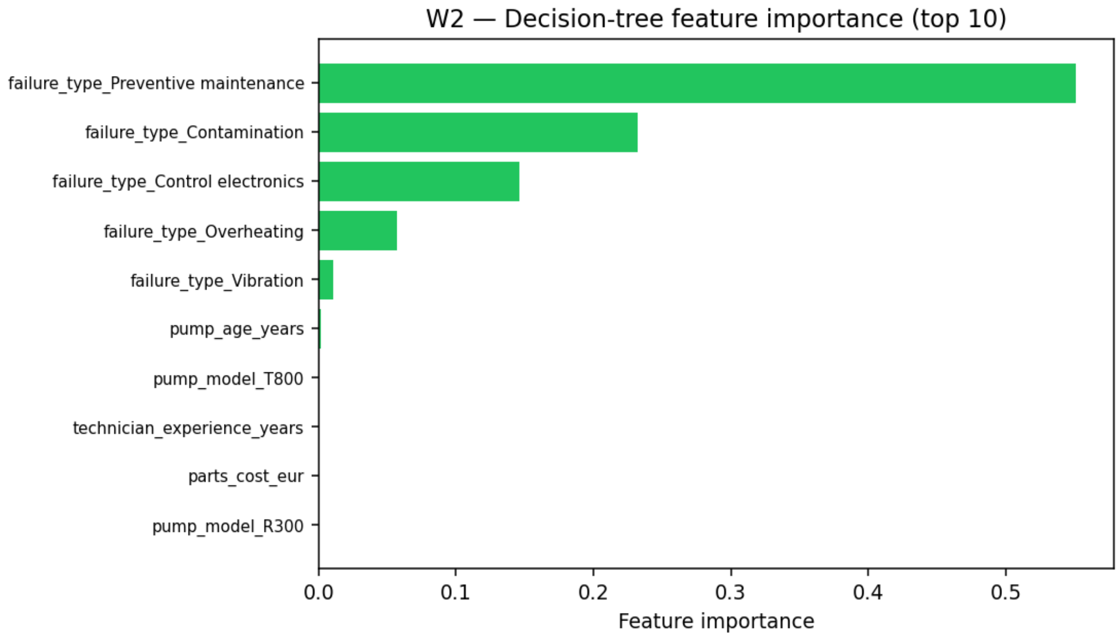
Frameworks

scikit-learn DecisionTreeRegressor · CART (Breiman et al. 1984)

[8] Breiman et al. 1984 [9] Rudin 2019

Usable intake-time ETAs — publish as a tailored range





Model Reliability (R^2)	Avg. Prediction Error	Risk Buffer Margin (RMSE)	Repairs Analyzed
92.7%	1.1 Days	2.2 Days	10000 Cases

Management action

- Publish a standard ±1-day delivery window on the receipt for predictable repairs (e.g. Overheating).
- Route highly unpredictable repairs ('Control electronics' and 'Seal/Leak') to senior staff and quote a wider ±4-day window.
- Standardize repair procedures for high-variance failure categories to systematically reduce scheduling errors.

[10] Jang et al. 2021 [11] Ding et al. 2022 · Acela

Which repairs deserve a closer pre-QA look?

Business question

Before signing off a repair, operations needs an automated alert system. The model must calculate failure risk at intake and trigger a review flag, balancing QA team capacity against quality goals.

Data used

- Target: QA Inspection Failure Flag (historical rate ≈13%)
- Predictors: Pump age, technician experience, estimated parts cost
- Job Info: Specific pump model, complexity class, failure type
- Preparation: Stratified data split to keep failure rates fully representative

Decision Tree Classifier + fixed threshold

- Calculate failure risk probability -> trigger alert at 30% risk
- Clear trigger threshold = highly governable dashboard flag
- Evaluated by: failures successfully caught vs review workload

Fixed hyperparameters

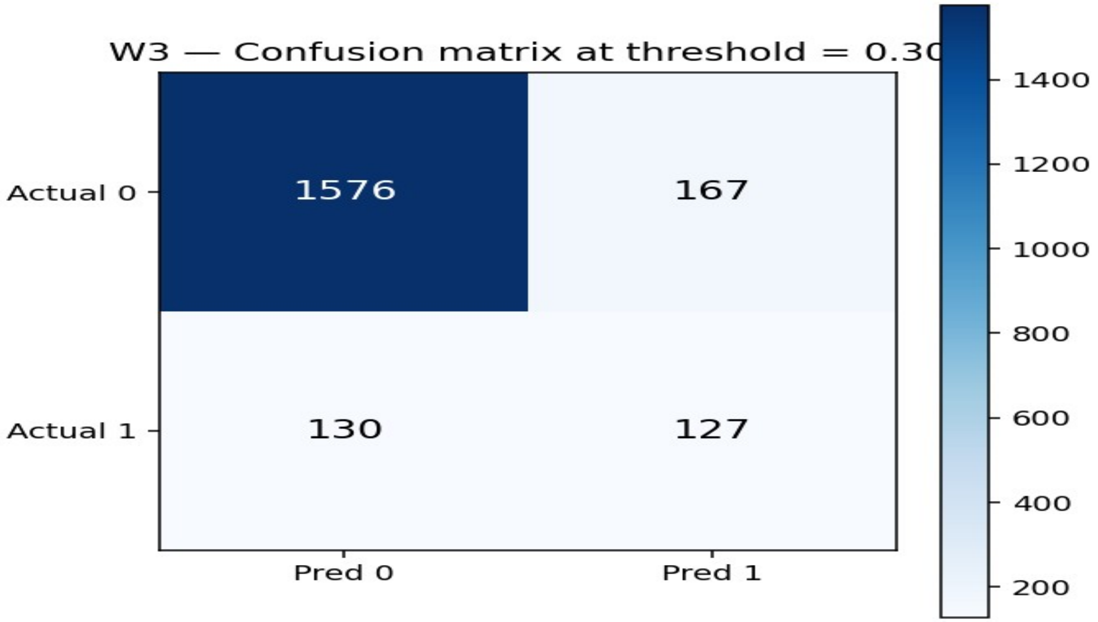
- max_depth = 6 ; min_samples_leaf = 20 ; random_state = 7
- Alert Trigger Threshold = 30% (optimized for shop review capacity)
- Stratification active to preserve natural 13% failure rate in evaluation

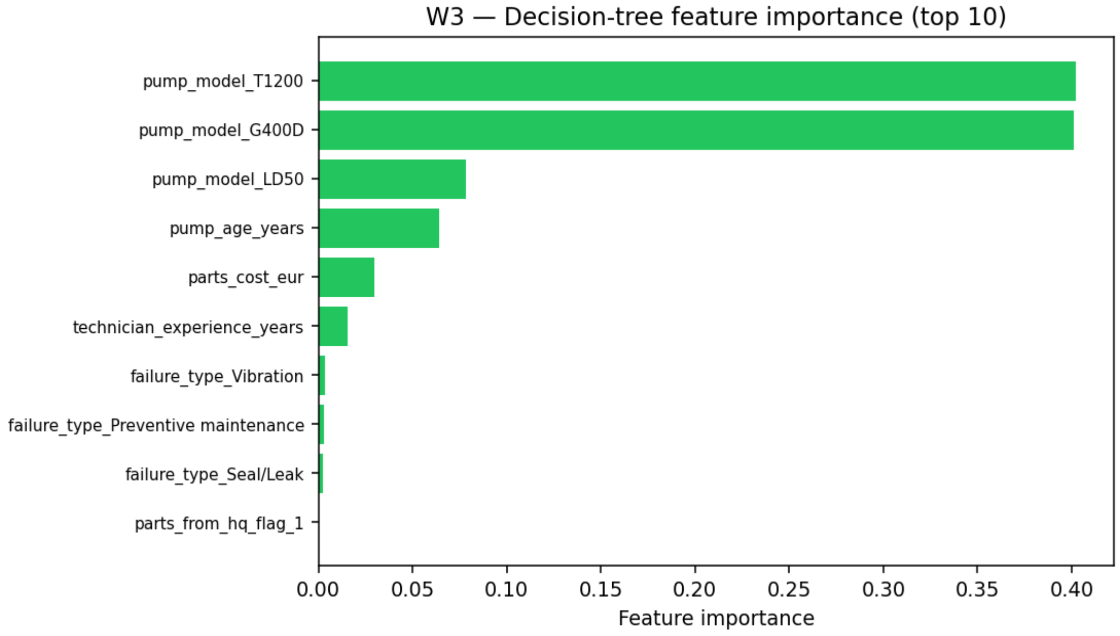
Frameworks

scikit-learn DecisionTreeClassifier · Imbalanced classification metrics

[8] Breiman et al. 1984 [9] Rudin 2019

Triage high-risk pump models 'T1200' and 'G400D'





Failures Caught (Recall)	Alert Accuracy (Precision)	Overall Triage Accuracy	Risk Trigger Threshold
49%	43%	85%	30%

Management action

- Route flagged jobs and high-risk pump models 'T1200' and 'G400D' through a mandatory senior tech pre-QA check.
- Use the risk flag to intercept 49% of all pre-QA failures early, keeping extra review workload below 15%.
- Run low-risk pump models ('G200D', 'R300') on an automated fast-track QA queue to maximize throughput.

[12] Zhang et al. 2023 [13] Sonawani & Mukhopadhyay 2013

CONCLUSION

Three concrete recommendations for VacuTech management

W1 · Parts Stocking

Optimize regional warehouse stocking by holding the top 26 critical add-on parts. Deploy the 306 high-confidence parts bundling rules to ensure technicians complete maintenance visits in a single trip.

CRM Value: Customer Development — Boost first-visit completion rates.

W2 · Smart Scheduling

Provide reliable repair windows based on our 93% reliable duration model. Quote a standard ±1-day window for predictable repairs, and route unpredictable categories ('Control electronics' and 'Seal/Leak') to senior staff with a wider ±4-day buffer.

CRM Value: Customer Retention — Fewer status check calls.

W3 · Quality Triage

Automatically flag high-risk jobs at intake, focusing on vulnerable pump models like 'T1200' and 'G400D'. This triage check will intercept 49% of all potential quality issues early with a quick, senior technician review.

CRM Value: Customer Loyalty — Prevent costly repeat-failure cycles.

Why interpretable models? Stakeholders (planners, advisors, QA supervisors) must act on the output. A shallow, auditable model that they can read and override is preferable to a high-accuracy black box they cannot challenge[9-2].

[9] Rudin 2019 [2] CRISP-DM Consortium 2000

Peer-reviewed sources and frameworks

[1] Shearer, C. (2000). The CRISP-DM model: The new blueprint for data mining. Journal of Data Warehousing, 5(4), 13-22. link

[2] CRISP-DM Consortium (2000). CRISP-DM 1.0: Step-by-step data mining guide. link

[3] Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules in large databases. Proc. VLDB '94, 487-499. link

[4] Brin, S., Motwani, R., Ullman, J. D., & Tsur, S. (1997). Dynamic itemset counting and implication rules for market basket data. Proc. ACM SIGMOD, 255-276. link

[5] Mundler, N. (2019). Association rule mining and itemset-correlation based variants. arXiv:1907.09535. link

[6] Wang, J., Pan, X., Wang, L., & Wei, W. (2018). Method of spare parts prediction models evaluation based on grey comprehensive correlation degree and association rules mining: A case study in aviation. Mathematical Problems in Engineering, 2018, 2643405. link

[7] Didriksen, S. K., Sigsgaard, K. W., Mortensen, N. H., & Jespersen, C. B. (2026). Assigning spare parts management decision-making strategies: A holistic portfolio classification methodology. Applied Sciences, 16(4), 1961. link

[8] Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). Classification and Regression Trees. Chapman & Hall. link

[9] Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, 1, 206-215. link

[10] Jang, J., Nana, D., Hochschild, J., & de Lorenzo, J. V. H. (2021). Predicting breakdown risk based on historical maintenance data for Air Force ground vehicles. arXiv:2112.13922. link

[11] Ding, Y., Gao, A., Ryden, T., et al. (2022). Acela: Predictable datacenter-level maintenance job scheduling. arXiv:2212.05155. link

[12] Zhang, Y., Gao, Z., Sun, J., & Liu, L. (2023). Machine-learning algorithms for process condition data-based inclusion prediction in continuous-casting process. Sensors, 23(15), 6719. link

[13] Sonawani, S., & Mukhopadhyay, D. (2013). A decision tree approach to classify web services using quality parameters. arXiv:1311.6240. link

[14] Ngai, E. W. T., Xiu, L., & Chau, D. C. K. (2009). Application of data mining techniques in customer relationship management: A literature review and classification. Expert Systems with Applications, 36(2), 2592-2602. link

Frameworks used

- scikit-learn (Decision Trees)
- mlxtend (Apriori, association rules)
- pandas (data manipulation)
- Jupyter (CRISP-DM notebooks)
- CRISP-DM 1.0 guide

See ACADEMIC_METHOD_JUSTIFICATION.md for full per-W argumentation.